

# Introduction

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The finite element method is arguably the principle extant numerical technique for solving partial differential equations, particularly over non-trivial domain geometries, and can be interpreted to contain a number of other popular approaches (finite volume, finite difference) as special cases.

### 1. The functional-analytic viewpoint on PDEs

A **linear space** or **vector space**  $V$  is a collection of objects that are closed under linear combination, i.e. if  $x, y$  are in  $V$  and  $\alpha, \beta$  are real or complex numbers, then  $\alpha x + \beta y$  is also in  $V$ . The classic examples are the real and complex Euclidean spaces  $\mathbb{R}^n$  and  $\mathbb{C}^n$ .

## Geometric Structure

A typical choice of inner product for a function space might be

$$\langle f, g \rangle := \int_{\Omega} fg dx, \quad (1)$$

$$(f, g) := \int_{\Omega} fg + \nabla f \cdot \nabla g dx, \quad (2)$$

where  $f, g$  are defined on the same domain  $\Omega$  and  $\nabla$  denotes the gradient operator. An inner product induces a norm, e.g.

$$\|f\| := \sqrt{\langle f, f \rangle} \quad (3)$$

defines a norm on many common function spaces.<sup>1</sup>

<sup>1</sup> $f, g$  may have to be regarded as equivalence classes of functions in order to satisfy the reflexivity condition but this subtlety can be safely ignored in this discussion

# Topological Structure

A **Hilbert space** is an inner product space which is complete with respect to the norm induced by the inner product. A function space might fail to be complete with respect to one norm, even if it is complete when equipped with another.

The **dual space** of a normed space  $V$ , denoted  $V^*$ , is the collection of scalar-valued, continuous linear functions defined on  $V$ .

## Example

In the case of  $\mathbb{R}^n$ , any scalar-valued linear mapping  $x^*$  acts on vectors  $x$  via

$$x^*(x) = a_1x_1 + a_2x_2 + \dots + a_nx_n \quad (4)$$

and so the dual can be identified with row vectors  $[a_1 a_2 \dots a_n]$

I will refer to elements of  $V^*$  as **functionals**.

# Dual Spaces

In an inner product space  $(V, [\cdot, \cdot])$  one way to define a functional is by a duality pairing: for  $v$  in  $V$ , we have that  $[v, \cdot] =: v^*$  is in  $V^*$ . That is,

$$v^*(u) = [v, u] \tag{5}$$

for  $u$  in  $V$ .

## Dual Space Representation

This observation might motivate one to ask: are *all* linear functionals representable in this way? In other words, if  $R_V: V \rightarrow V^*$  is defined by

$$R_V(v) \mapsto [v, \cdot] \tag{6}$$

invertible?

## Theorem (Riesz Representation theorem)

Suppose that  $(H, [\cdot, \cdot])$  is a (real) Hilbert space. Then  $R_H$  is a linear isomorphism.

In particular,  $R_H$  is invertible.

## Variational Form

The point of this whole discussion is that in order to make progress on some PDE, say

$$A[u] = 0, \tag{7}$$

$$\tag{8}$$

if we can establish a corresponding problem in the dual of some Hilbert space  $H$ , say

$$R_H(u) = v^*. \tag{9}$$

$$\tag{10}$$

for some  $v^*$  in  $H^*$ , we may conclude that  $u = R_H^{-1}(v^*)$ .



## Example

Consider the PDE

$$-\Delta u = f \quad \text{in } \Omega \quad (11)$$

$$u = 0 \quad \text{on } \partial\Omega \quad (12)$$

For some  $\Omega \subset \mathbb{R}^n$ .<sup>2</sup>

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<sup>2</sup>This PDE is called the Poisson equation where the operator  $\Delta = \frac{\partial}{\partial x_1^2} + \dots + \frac{\partial}{\partial x_n^2}$ .

## Theorem (The Lax-Milgram theorem)

Suppose that  $a$  is a bilinear form,  $L$  is a linear form defined on a Hilbert space  $V$ . If  $a$  is

1. **Bounded**, i.e.  $a(u, v) \leq C_B \|u\| \|v\|$  for all  $u, v$  in  $V$ ,
2. **Coercive**, i.e.  $a(v, v) \geq C_C \|v\|^2$ ,

then there is a unique solution  $u$  to the variational problem

$$a(u, v) = L(v) \quad \text{for all } v \text{ in } V \quad (13)$$

and furthermore  $\|u\|$  can be bounded by the initial data  $L$ .

### 2. The finite element method

To develop a discrete problem, we instead consider a finite subspace

$$V^h = \text{span}\{\phi_1, \dots, \phi_N\} \subset V \quad (14)$$

## Discrete problem

Since  $a$  is assumed to be bilinear and  $L$  linear, if the coordinates of  $u^h$  are given by  $u^h = u_1\phi_1 + \dots + u_N\phi_N$ , then the discrete variational problem corresponds to the linear system

$$\sum_{j=1}^N u_j a(\phi_j, \phi_i) = L(\phi_i), \quad i = 1, \dots, N. \quad (16)$$

# Consistency

Presumably,  $V^h$  will be chosen to approximate  $V$  as  $N \rightarrow \infty$ . In the case where  $a$  is an inner product, if  $a(u, v^h) = L(v^h)$ , then clearly

$$a(u - u^h, v^h) = L(v^h) - L(v^h) = 0.$$

This means that the mapping  $u \mapsto u_h$  defines an orthogonal projection from  $V$  on the subspace  $V^h$ , and that the  $a$ -norm of the error  $u - u^h$  is minimal within  $V^h$ .

## Geometry setup

```
#FEniCS example
#| echo: true
from dolfinx import mesh, default_scalar_type
from dolfinx.fem import FunctionSpace, Constant, locate_dof
from dolfinx.fem.petsc import LinearProblem
from mpi4py import MPI
import numpy as np
import ufl

#mesh size
h = 20
#create the domain
#comm argument is related to multiprocessing which isn't re
domain = mesh.create_unit_square(comm = MPI.COMM_WORLD,
                                nx = h,
                                ny = h)

#test/trial function space V
#1-dimensional function space in 2D
```

# Variational Problem Formulation

```
#define trial and test function

u = ufl.TrialFunction(V)
v = ufl.TestFunction(V)

#define forcing term
f = Function(V)
f_expr = lambda x: 2 * np.sin(2 * np.pi * x[1]) * (1 - 4 *
f.interpolate(f_expr )

A = (ufl.dot(ufl.grad(v), ufl.grad(u))) * ufl.dx
L = v * f * ufl.dx
```

## Assembly and linear solver

```
problem = LinearProblem(A, L, bcs=[boundary_condition])  
uh = problem.solve()
```

# Extensions

- ▶ Nonlinear problems can be solved by Newton-type iteration methods
- ▶ The assumptions of the Lax-Milgram Theorem can be weakened
- ▶ Many alternative types of spatial discretizations (e.g. different shapes or nonconforming meshes)
- ▶ Alternative families of elements may be useful in certain circumstances
- ▶ Other types of variational problems can be considered